

Reviewer Pack — Minimal Rank of Geometric Norms over Boolean Functions vs De...

Entry ded464fb6192 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-25 16:48:14 UTC

Minimal Rank of Geometric Norms over Boolean Functions vs Decision Tree Path Complexity

Entry ID: ded464fb6192 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 16:48:14 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Geometric Norms) **Field B** (complexity object): Complexity Theory: Decision Tree Path Complexity

Statement:

{'text': 'For every Boolean function f with n variables, let N_f be the minimal rank of a nondegenerate Hermitian form that upper-bounds the modulus of any Taylor coefficient of f . Then, for all instances of size $n \leq 40$, the decision tree path complexity $D(f)$ is at most $O(N_f^{\{1/2\}} \log^2 n)$.', 'counterexample': 'A Boolean function f with a high-degree polynomial form and a Hermitian form with rank significantly lower than $D(f)^2$.']}

Rationale (proposer's reasoning):

{'text': 'Geometric norms provide a tool for studying the arithmetic structure of functions, which can potentially reveal hidden complexity-theoretic invariants. This conjecture links geometric properties of functions to decision tree complexity, offering a new angle for understanding computational hardness.'}

Taxonomy category: Geometric_Norms_x_Complexity_Theory (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 9a604952b0b49d00

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For the conjecture that $D(f) \leq O(N_f^{\{1/2\}} \log^2 n)$, results will be considered supported if the median of $N_f/D(f)^{\{1/2\}}$ for all $n \leq 40$ is less than or equal to $\log^2 n$ with a p-value < 0.05 , and falsified if any seed produces a ratio greater than $\log^2 n$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Rank Geometric Norms AND Decision Tree Path Complexity - Hermitian Form Rank Boolean Functions related to Decision Tree Path Complexity - Taylor Coefficient Modulus Upper Bound and Decision Tree Path Complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
from fractions import Fraction
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def taylor_coefficients(f, n):
        # Placeholder for actual Taylor coefficient computation
        return [sum(f[i] * (x ** i) for x in range(2)) for i in range(n + 1)]

    def hermitian_form_rank(coeffs):
        # Placeholder for actual Hermitian form rank computation
        return len(coeffs)

    def decision_tree_path_complexity(f, n):
        # Placeholder for actual decision tree path complexity computation
        return sum(1 for coeff in coeffs if coeff != 0)

    n = random.randint(5, 40)
    f = generate_boolean_function(n)
    coeffs = taylor_coefficients(f, n)
    N_f = hermitian_form_rank(coeffs)
    D_f = decision_tree_path_complexity(f, n)

    if D_f == 0:
        return {
            "metric_name": "N_f / D(f)^{1/2}",
            "metric_value": float('inf'),
            "instances_tested": 1,
```

```

        "conjecture_holds": False,
        "counterexample": "D(f) is zero, making the ratio undefined."
    }

ratio = Fraction(N_f, D_f**0.5).limit_denominator()
return {
    "metric_name": "N_f / D(f)^{1/2}",
    "metric_value": float(ratio),
    "instances_tested": 1,
    "conjecture_holds": ratio <= math.log2(n)**2,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all("conjecture_holds" in r and r["conjecture_holds"] for r in results):
        support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)
        print(f"RESULT: SUPPORTED mean={sum(r['metric_value'] for r in results) / len(results)} st")
    elif any("counterexample" in r and r["counterexample"] for r in results):
        first_failing_seed = next(s for s, r in zip(seeds, results) if "counterexample" in r and r["counterexample"])
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing")
    else:
        print("RESULT: INCONCLUSIVE")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, making it impossible to evaluate the conjecture's support or falsification. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11446
2	preregistration	ollama_remote	glm4:latest	0	0	5963
3	novelty	ollama_remote	glm4:latest	0	0	4607
4	novelty	ollama_remote	glm4:latest	0	0	6552
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15048
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7921
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11226
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7960
9	judge	ollama_remote	glm4:latest	0	0	15980

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 86702 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/ded464fb6192.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/ded464fb6192.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/ded464fb6192.tar.gz` (if generated)